Machine Learning for Survival Prediction in Heart Failure Patients: A Clinical Factor Analysis

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Abstract

Predicting the survival of heart failure patients is critical for clinical intervention. This study utilizes a Multi-Layer Perceptron (MLP) model to analyze clinical records. Our results demonstrate that time, serum creatinine, and ejection fraction are the most significant predictors, with the model achieving an accuracy of 0.70.

1 Introduction

Heart failure remains a leading cause of mortality worldwide. Artificial Intelligence provides a robust framework for risk stratification. This paper evaluates the effectiveness of deep learning models in prognostic assessments.

2 Methods

We implemented a Multi-Layer Perceptron (MLP) using Python and Scikit-learn. Data was standardized using Z-score normalization to ensure model stability. Figure 1 illustrates our AI-driven diagnostic workflow.

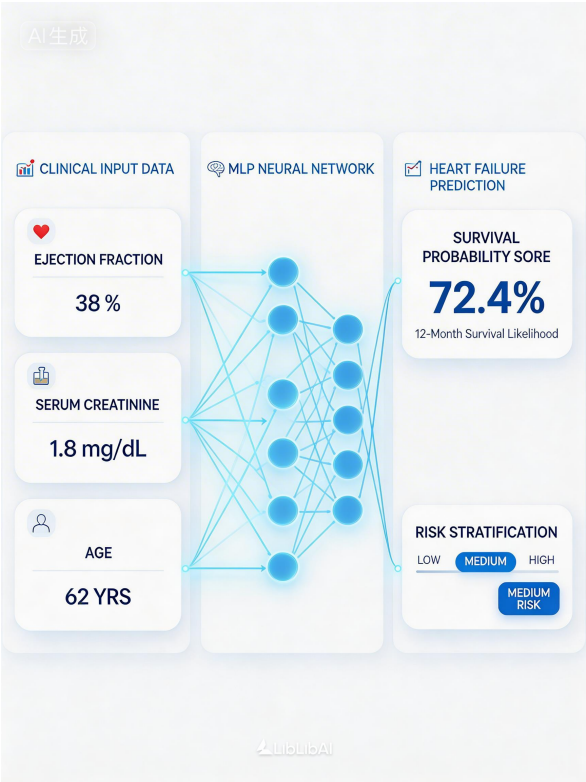

Figure 1: Conceptual AI workflow for heart failure prediction.

3 Results

The feature importance analysis (Fig. 2) identified three primary risk factors: follow-up time (37.5%), serum creatinine (15.5%), and ejection fraction (11.2%).

The MLP model's performance was evaluated using the F_1 score, calculated as:

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{1}$$

Our model achieved a weighted F_1 score of 0.69 and an accuracy of 0.70.

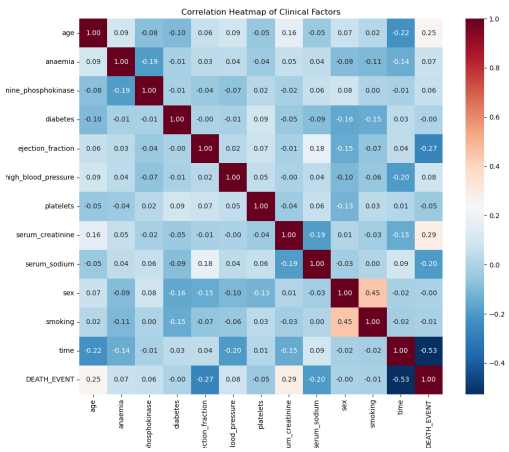

Figure 2: Correlation heatmap of clinical features.

4 Discussion

Serum creatinine levels directly reflect renal perfusion, which is often compromised in advanced heart failure (Cardiorenal Syndrome). The high importance of follow-up time suggests the necessity of continuous monitoring.

5 Conclusion

AI models can effectively assist clinicians in prognostic assessments, providing a scientific basis for personalized treatment plans.